

Instructions

Solve the following problems using the properties of logarithmic functions. Show all steps clearly, and simplify your answers as much as possible.

Problems

Problem 1. *Simplify the expression using logarithmic properties:*

$$\log_5(25) + \log_5(5)$$

Problem 2. *Solve for x using the properties of logarithms:*

$$\log_2(x) + \log_2(8) = 4$$

Problem 3. *Simplify:*

$$\log_3(81) - \log_3(9)$$

Problem 4. *Use the change of base formula to evaluate the logarithm:*

$$\log_4(16)$$

Problem 5. *Simplify the expression:*

$$\frac{1}{3} \log_2(8) + \frac{1}{2} \log_2(4)$$

Problem 6. *Solve for y in the equation:*

$$\log_{10}(y) = \log_{10}(5) + \log_{10}(2)$$

Problem 7. *Express the logarithmic expression as a sum or difference of logarithms:*

$$\log_b \left(\frac{x^2 \cdot y^3}{z} \right)$$

Problem 8. *Use the properties of logarithms to simplify:*

$$\log_7(7^4) - 2 \log_7(7)$$

Problem 9. *Solve the logarithmic equation for x :*

$$2 \log_3(x) = \log_3(9)$$

Problem 10. *Express the following logarithmic expression using natural log:*

$$\log_5(50)$$

Problem 11. *Simplify $10^{-\log_{10}(2x)}$*

Problem 12. *Simplify $(2^{\log_2(x)})^{-1}$*

Problem 13. *Simplify $e^{-3\ln(2)}$*

Problem 14. *Simplify $e^{-\frac{1}{2}\ln(16)}$*

Problem 15. *Solve the logarithmic equation for x :*

$$\log_3(x) + \log_3(x - 2) = 1$$

Problem 16. *Solve the logarithmic equation for x :*

$$\ln(x) - \ln(x - 3) = \ln(2)$$